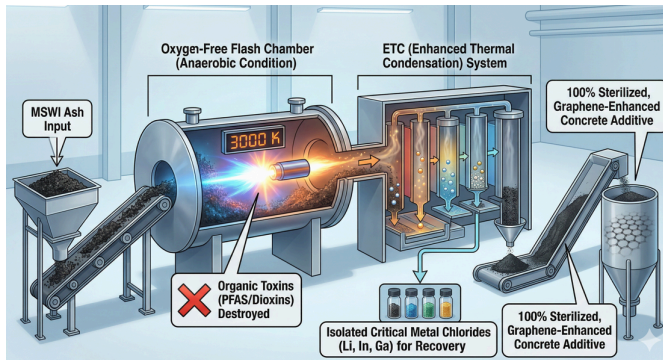


Project Sp-ARC: TRL 5/6 Tactical Urban Mining & Ash Neutralization

Operational Impact

- Strategic Resource Independence: Secures domestic supplies of Lithium, Indium, and Gallium from municipal waste streams to bypass foreign supply chains.
- Naval Workforce Integration: Capitalizes on existing DoD technical training: current WTE operational protocols map to ex-Navy technician skill sets.
- Base Resilience and Logistics: Decentralizing metal recovery at regional facilities eliminates heavy transportation logistics and local environmental liabilities.



Status and Capability

- Current Capability Gap: Legacy MSWI ash management relies on toxic chemical leaching or hazardous landfilling, losing critical minerals to a silica matrix.
- Technology Readiness Level: The Sp-ARC project transitions Gen-1 Flash Joule Heating from lab environments to a TRL 5/6 operational pilot in Spokane.
- Strategic Alignment: Project supports the DoD SAFiRE mandate for operational energy efficiency and domestic supply chain security.
- Administrative Readiness: Advanced Recovery Coalition holds an active UEI (HXYTRTNMGU6) and is prepared for immediate federal obligation.

Technical Approach

- System Mechanism: Deployment of a 1-TPD FJH reactor with Electro-Thermal Chlorination (ETC) to process pre-milled, electroconductive MSWI ash.
- Thermodynamic Neutralization: Millisecond thermal shock to 3000 K in an oxygen-free environment achieves measured destruction of organic toxins like PFAS.
- Fractional Vaporization: Pure chlorine gas injection drives the recovery of 75 percent to 85 percent pure critical metal chlorides (Li, In, Ga).
- Beneficial Byproduct: The remaining 15 percent inert material serves as a sterilized, graphene-like strengthening additive for MILCON-grade concrete.

Schedule and Cost

- Phase 1 Timeline: 12-month cycle for site preparation, hardware commissioning, and third-party material characterization.
- Equipment and Commissioning: \$1.4M (46.6 percent) for Gen-1 FJH unit lease, modular integration, and baseline 3000 K calibration.
- Mandatory EWEV&DM: \$500K (16.7 percent) for a specialized sensor array to monitor electrical load and grid impact for DoD visibility.
- Validation and Admin: \$1.1M (36.7 percent) for [PI Name] led lab verification, municipal MSA execution, and SAM.gov compliance.